

A Dual-Branch Dynamic Graph Convolution Based Adaptive TransFormer Feature Fusion Network for EEG Emotion Recognition

Mingyi Sun, Weigang Cui , Shuyue Yu , Hongbin Han, Bin Hu , *Member, IEEE*, and Yang Li 

Abstract—Electroencephalograph (EEG) emotion recognition plays an important role in the brain-computer interface (BCI) field. However, most of recent methods adopted shallow graph neural networks using a single temporal feature, leading to the limited emotion classification performance. Furthermore, the existing methods generally ignore the individual divergence between different subjects, resulting in poor transfer performance. To address these deficiencies, we propose a dual-branch dynamic graph convolution based adaptive transformer feature fusion network with adapter-finetuned transfer learning (DBGC-ATFFNet-AFTL) for EEG emotion recognition. Specifically, a dual-branch graph convolution network (DBGCN) is firstly designed to effectively capture the temporal and spectral characterizations of EEG simultaneously. Second, the adaptive Transformer feature fusion network (ATFFNet) is conducted by integrating the obtained feature maps with the channel-weight unit, leading to significant difference between different channels. Finally, the adapter-finetuned transfer learning method (AFTL) is applied in cross-subject emotion recognition, which proves to be parameter-efficient with few samples of the target subject. The competitive experimental results on three datasets have shown that our proposed method achieves the promising emotion classification performance compared with the state-of-the-art methods. The code of our proposed method will be available at: <https://github.com/smy17/DANet>.

Index Terms—EEG, emotion recognition, graph neural network, Transformer, transfer learning

1 INTRODUCTION

THE brain computer interface (BCI) links us with computers in an innovative way [1]. Emotions play an important role in the process of communication. As one of the most significant interdisciplinary fields, researches on affective

Brain-Computer Interactions (aBCIs) have received increasing attention [2], [3]. Traditional input signals for affective computing include audio, text and video (facial expressions, body movements, etc.) [4]. As the rise of metaverse, researchers showed great interests in the role of physiological signals, for example, EEG, EOG and ECG [5]. In recent years, there are plenty of researches on neuroscience demonstrating that the change of EEG signals is strongly associated with emotion status [6], [7]. Apart from its reliability, the flexible collection of EEG signals has made it possible to conduct real-time monitoring of emotions [8], [9]. Thus, the EEG based pattern recognition and correct decoding are of great necessity in the affective computing area.

In order to realize EEG-based emotion recognition, many classical machine learning algorithms have been proposed to classify the hand-crafted features from original EEG signals. For example, Murugappan et al. [10] and Reuderink et al. [11] adopted shallow models to classify the time-frequency domain features. Differential entropy (DE) features of EEG segments were computed and classified by the k-Nearest Neighbor (kNN) method [12], which avoided heavy computational cost of raw EEG signals. Zheng et al. [13] further introduced a deep belief network and constructed the three-type emotion classification model with the preprocessed DE features as inputs. A concatenated feature map, which was obtained from differential entropy and power spectral density (PSD), was used for further detection of emotion status [14]. However, these methods above just processed the obtained feature map in the single shallow network without taking full advantage of temporal and

- Mingyi Sun is with the Department of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China. E-mail: sunmingyi@buaa.edu.cn.
- Weigang Cui is with the School of Engineering Medicine, Beihang University, Beijing 100191, China. E-mail: cwg1994@buaa.edu.cn.
- Shuyue Yu is with Beijing Aerospace Measurement & Control Technology Co. Ltd, Beijing 100024, China. E-mail: ysy_ivy@163.com.
- Hongbin Han is with the Institute of Medical Technology, Peking University Health Science Center, Peking University, Beijing 100191, China, and also with the Beijing Key Laboratory of Magnetic Resonance Imaging Technology, Beijing 100191, China. E-mail: hanhongbin@bjmu.edu.cn.
- Bin Hu is with the Gansu Provincial Key Laboratory of Wearable Computing, School of Information Science and Engineering, Lanzhou University, Lanzhou, Gansu 730000, China. E-mail: bh@lzu.edu.cn.
- Yang Li is with the Beijing Advanced Innovation Center for Big Data and Brain Computing, Department of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China. E-mail: liyang@buaa.edu.cn.

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(Corresponding author: Yang Li.)

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spectral features simultaneously, which results in the insufficient extraction of EEG emotional information.

In addition, recent studies have demonstrated that deep learning methods are able to learn more discriminative features from data automatically [15]. Many researchers have started to explore high-level information in EEG emotion recognition with graph neural networks [16] and then fuse the extracted feature maps with shallow 2D convolution. For instance, Song et al. [17] applied the graph neural network to classify the DE features, which efficiently modeled the connection between different EEG channels with a dynamic adjacent matrix. In order to simultaneously take the distribution of brain regions into account, a novel feature representation of EEG features was discussed [18] and 4D convolution was applied to extract high-level feature maps. However, conventional CNNs can only focus on local spatial features of the brain network, which led to the loss of patterns in higher dimensional space. In order to avoid this limitation, the attention units were applied with LSTM to improve the invariance ability against the emotional intensity fluctuation and automatically adjust the weights of channels [35], which may provide a solution to efficiently fusing the obtained feature maps from GNNs.

Some recent studies started to introduce cross-subject experiments in order to enable rapid application of emotion recognition in BCIs. Although the classical subject-dependent methods [18] have been proved to show an outstanding performance in emotion recognition, the risk of overfitting and dependence on the amount of data limit its flexible application. In order to overcome these problems, Song et al. [17] and Li et al. [35] applied the leave-one-subject-out method to transfer the general emotion pattern of source subjects to the target subject. Although these methods achieved promising classification results, non-negligible divergence between different individuals and the simple transferring without adaption maybe leads to misjudgment of the pretrained model, which results in poor classification performance on the target subject.

To address the issues above, in this article, we propose a novel dual-branch dynamic graph convolution based adaptive transformer feature fusion network with adapter-finetuned transfer learning, namely, DBGC-ATFFNet-AFTL for EEG emotion recognition. First, both differential entropy and power spectral density features of each EEG segment are computed and the dual-branch graph dynamic convolution (DBGC) network is designed to capture deeper temporal and spectral features in different frequency bands from dual branches respectively. Second, the Adaptive Transformer feature fusion Network (ATFFNet) is adopted to apply self-attention mechanism on different kinds of feature maps in consideration of the channel connection, in order to effectively capture the global pattern of the emotion status. Moreover, the adapter-finetuned transfer learning (AFTL) algorithm is proposed to efficiently avoid the overfitting problem caused by insufficient samples of target subject through finetuning the Adapter modules. The proposed DBGC-ATFFNet-AFTL method is evaluated on three public datasets and gains promising performance compared with the state-of-the-art methods, demonstrating its efficacy in EEG emotion recognition.

Main contributions of this paper are summarized as follows:

- 1) We propose a novel DBGC-ATFFNet-AFTL method for EEG emotion recognition, which integrates high-level features with dual branches into the deep learning network. The proposed DBGC-ATFFNet-AFTL method performs more accurately and efficiently on emotion classification than the widely used dynamic Graph Neural Network.
- 2) A dual-branch dynamic graph convolution block is developed to acquire the temporal and spectral characteristics by dual branches, which overcomes the weakness of insufficient emotional information extraction with a single encoding path.
- 3) We design an adaptive transformer feature fusion network to implement the fusion of high-level temporal and spectral features simultaneously, efficiently associating the spatial distribution of EEG channels with deeply encoded emotion characteristics, and thus boosts the classification performance.
- 4) We propose an adapter-finetuned transfer learning algorithm to realize the rapid cross-subject EEG emotion recognition through finetuning the Adapter modules. It can effectively avoid the overfitting problem brought by the subject-dependent method and show an outstanding performance with quite a small number of trainable parameters.

2 METHODOLOGY

The overall architecture of our proposed DBGC-ATFFNet-AFTL is outlined in Fig. 1, and summarized as follows. For the design of our proposed model, differential entropy (DE) and power spectral density (PSD) of EEG segments are first calculated by inverse fast Fourier transform (IFFT) and short time Fourier transform (STFT), and then the DBGC module captures both the temporal DE and spectral PSD information of EEG signals by using dual branches of graph convolution; second, the ATFFNet effectively fuses the obtained feature maps, which consists of multi-head self-attention mechanism and subject-adaptive unit (SAU); finally, the classification block gives the final recognition results. Additionally, the pipeline of our proposed adapter-finetuned transfer learning is given as follows: 1) We firstly divide the whole dataset into two parts: the i -th subject is the target subject, which is randomly selected, while the rest ($N-1$) subjects are the source subjects. Specifically, half samples of the target subject are used for finetuning while the other half samples from the i -th subject are applied to evaluate the classification performance; 2) We then pretrain the proposed model on all samples of source subjects and update all the trainable parameters; 3) Based on training samples of the target subject, we only finetune the parameters of Adapter, which is embedded in the SAU, to bridge the gaps between the target subject and source subjects. 4) Finally, the well-trained model is evaluated on test samples of the target subject. In the following three subsections, we discuss in detail the specific implementation of the proposed innovation modules.

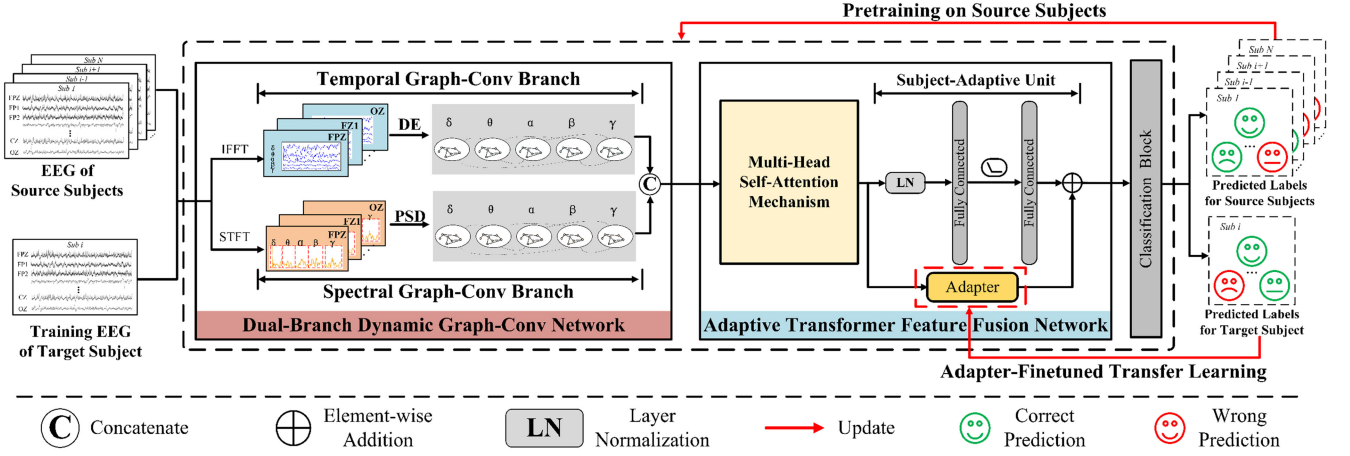


Fig. 1. The pipeline of the proposed DBGC-ATFFNet-AFTL method.

2.1 Dual-Branch Dynamic Graph Convolution Network

The original EEG signals are defined as $E = \{(X_i, y_i) | i = 1, 2, \dots, K\}$, where $X_i \in R^{C \times S}$ is a two-dimension array representing the i -th EEG trail with C channels and S samples. K is the total number of EEG signal trails. y_i is the corresponding label of X_i and takes its value from label set L including M classes in an emotion recognition task. For example, the label set for the discrete datasets (i.e., SEED) consists of the explicit emotion statuses: $L = \{l_1 = \text{"neutral"}, l_2 = \text{"happy"}, l_3 = \text{"sad"}\}$.

As previous works [13], [20], we firstly split the original EEG signals with a T s-long window without overlapping, and it has been proved that real-time emotion recognition can be approximately realized with emotional information largely preserved when T is set 1 [17]. Each segment is assigned with the same label as the original EEG signals.

According to the experiment results in recent studies [17], [18], differential entropy (DE) and power spectral density (PSD) features have been proved to achieve a promising performance in depicting the emotional fluctuation. However, for different types of emotional stimulation, the two features make a different contribution to the final recognition result. Under video stimulation (i.e., SEED dataset), for example, DE features perform better in identifying the emotion of subjects [12], while PSD features prove to have an outstanding result under music stimulation (i.e., DEAP dataset) [19]. Therefore, we choose both DE and PSD as the basic input features for our proposed model.

The two types of features of all EEG channels are computed according to the five frequency bands [20] (i.e., δ [1~4 Hz], θ [4~8 Hz], α [8~14 Hz], β [14~31 Hz] and γ [31~51Hz]), both of which have been proven to be effective for emotion recognition [5]. The DE feature for Gaussian distribution is defined as follows [12]:

$$D(X) = - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \ln \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx$$

$$= \frac{1}{2} \ln 2\pi e \sigma^2 \quad (1)$$

where X denotes the Gaussian distribution $N(\mu, \sigma^2)$, x is a variable, π and e are constants, respectively.

The PSD features can be computed by using Short Time Fourier Transform (STFT), which is defined by [17]:

$$h_p(X) = E[x^2] \quad (2)$$

where x is formally a signal variable acquired from a certain frequency band on a certain EEG channel.

Therefore, we extract the DE feature tensor and PSD feature tensor, $F_D, F_P \in R^{C \times B}$ from each segment, where C is the number of EEG channels and B is the number of frequency bands, respectively. Since the above extracted features are relatively independent and fail to fully consider the effects of different brain regions on emotion, in this part, a DBGC module is designed to further explore deeper temporal and spectral features with channel relationship. The layout of the detailed DBGC is depicted in Fig. 2, which includes two synchronized Graph-Conv branches with a deeply encoded adjacent matrix.

Based on the previous extracted DE and PSD features, a dynamic adjacent matrix is proposed to model the connection among EEG channels [21], [22]. We firstly randomly initialize an adjacent matrix $A \in R^{C \times C}$, where the (i, j) -th element measures the coupling strengths between the i -th and j -th EEG channel. In this way, A shows that every channel is densely related with each other, taking direction and strength into account simultaneously. Then the matrix is encoded using Tanh nonlinearity to simulate the directional dependencies between different channels as follows:

$$\tilde{A}_{dd} = \sigma(W_2 \delta(W_1 \tilde{A})) \quad (3)$$

where $\tilde{A} \in R^{C \times C}$ is vectorized from A , $W_1 \in R^{(\frac{C \times C}{r}) \times (C \times C)}$ and $W_2 \in R^{(C \times C) \times (\frac{C \times C}{r})}$ are weight matrixes, $\delta(\cdot)$ and $\sigma(\cdot)$ are ELU and Tanh functions respectively, and r is the reduction ratio. Therefore, a dense adjacent matrix $A_{dd} \in R^{C \times C}$ is obtained by reshaping $\tilde{A}_{dd} \in R^{(C \times C) \times 1}$ into $R^{C \times C}$, where the (i, j) -th entry is learnable and reflects the directional dependency between the i -th and j -th EEG channel. Then we adopt a rectified linear unit (ReLU) to penal weak channel couplings and as a result, a non-negative adjacent matrix A_{ds} is achieved. Thus, $\mathcal{G}$ is defined as $\mathcal{G} = \{\mathcal{V}, F_D, F_P, A_{ds}\}$, where $\mathcal{V}$ is the vertex set with $|\mathcal{V}| = C$ nodes, node attributes

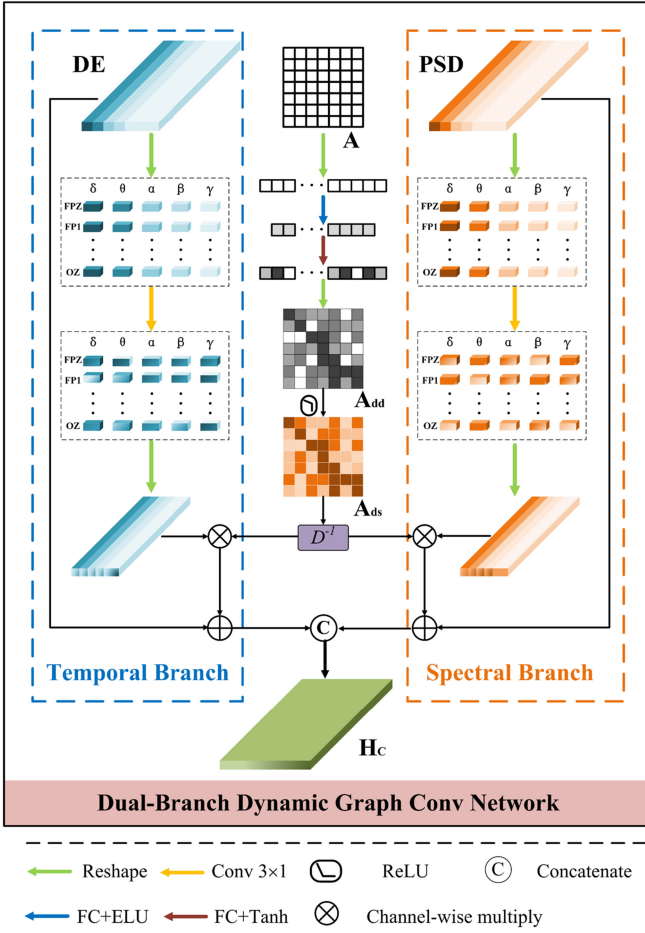


Fig. 2. The layout of dual-branch dynamic graph conv network.

are differential entropy and power spectral density features under different rhythms $F_D, F_P \in R^{C \times B}$, connections between nodes are determined by the sparse adjacent matrix $A_{ds} \in R^{C \times C}$.

Given two different types of EEG features as the inputs, one adjacent matrix is adopted to be shared in these two branches and we also introduce the residual architecture to avoid the gradient vanishing problem. Moreover, in order to reduce the heavy computation cost, the K order Chebyshev polynomials framework is not adopted in this paper as other GNN works do [17], instead, we operate graph convolution on the feature tensor extracted from multiple frequency bands. In this method, the hidden states of two branches can be computed via

$$\begin{aligned} H_D &= \delta_2(D^{-1}A_{ds}\delta_1(F_D W_{11})W_{12} + F_D) \\ H_P &= \delta_2(D^{-1}A_{ds}\delta_1(F_P W_{21})W_{22} + F_P) \end{aligned} \quad (4)$$

where $H_D, H_P \in R^{C \times B}$ are corresponding hidden states of DE and PSD features, $D^{ii} = \sum_j A_{ds}^{ij}$ is the degree matrix of A_{ds} , $W_{.1} \in R^{B \times B'}$ and $W_{.2} \in R^{B' \times B}$ are weight matrixes and B' is an adjustable hyperparameter, both $\delta_1(\cdot)$ and $\delta_2(\cdot)$ are ELU nonlinearity functions. We concatenate H_D and H_P above, and then obtain high-level features $H_C \in R^{C \times (B \times 2)}$ as the outputs.

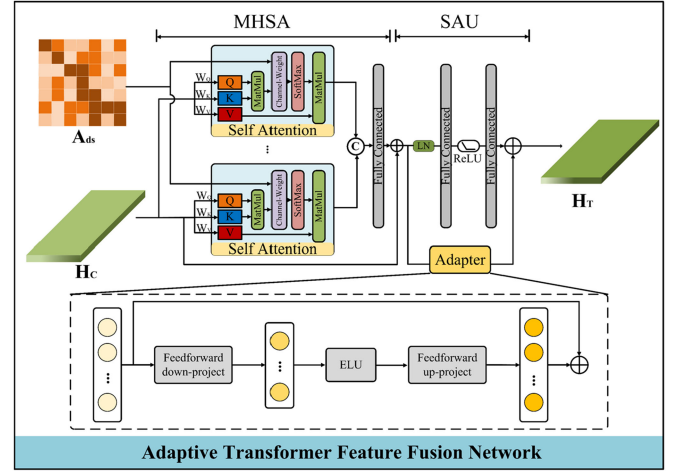


Fig. 3. The outline of adaptive transformer feature fusion network.

2.2 Adaptive Transformer Feature Fusion Network With Adapter-Finetuned Transfer Learning

From the aforementioned blocks, high-level features of EEG signals are generated from both the DE and PSD independently, which is regarded as the input feature H_C of the ATFF Network here. We can separate the whole block into two parts: the multi-head self-attention mechanism $MHSA(\cdot)$ and the subject-adaptive unit $SAU(\cdot)$. The output of the block H_T can be calculated via:

$$\begin{aligned} H'_T &= MHSA(LN(H)) + H_C \\ H_T &= SAU(LN(H'_T)) + H'_T \end{aligned} \quad (5)$$

where $LN(\cdot)$ denotes the layer normalization. As shown in Fig. 3, the $MHSA$ mainly consists of self-attention mechanisms and fully connected layer. For the self-attention mechanisms, there are totally three weight matrixes proposed: a query weight matrix $W^Q \in R^{(B \times 2) \times d_k}$, a key weight matrix $W^K \in R^{(B \times 2) \times d_k}$, and a value weight matrix $W^V \in R^{(B \times 2) \times d_v}$, where d_k and d_v are hyperparameters. Thus, the query Q , key K and value V can be defined as follows:

$$\begin{aligned} Q &= H_C W^Q \\ K &= H_C W^K \\ V &= H_C W^V \end{aligned} \quad (6)$$

As [23] points out, self-attention mechanism in the Transformer architecture is not sensitive to the positional information of the input sequences, therefore, positional embeddings should be added to construct the spatial relationship between the input elements. However, previous works [23], [24] only applied 1-Dimension absolutely positional embeddings, which is not optimal to the construction of brain channels due to omitting the dependent relationship from different channels. In order to take the dependent relationship between different EEG channels into full consideration, we introduce the adjacent matrix A_{ds} obtained from the above DBGCC module, which is defined as the Channel-Weight unit. As a result, the output of the self-attention mechanism Z can be calculated as follows:

$$Z = A_{ds} \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

Eq. (7) calculates the weight of the global spatial connection between all EEG channels and fuses different kinds of extracted features efficiently.

Moreover, we add the Adapter module to the subject-adaptive unit (SAU) apart from the common multi-layer perceptron [23] for the purpose of transfer learning. The architecture of adapter module is depicted in Fig. 3, which consists of a bottleneck including two feed-forward layers and an ELU activation unit. To make it more parameter-efficient, the adapter also contains a skip-connection. Adapter modules perform more general architectural modifications to make a pre-trained model suitable for the target subject, which involves adding a small number of new parameters. The detailed description of transfer learning is given in the next section. Finally, the fused feature map is obtained after processing the output of self-attention mechanism by means of the matrix multiplication and layer normalization.

2.3 Classification Block

The classification block in the proposed model is designed to give the final affective computing results based on the fused high-level features. We firstly flat all the feature maps into 1-Dimension vector and feed them into two fully connected layers. Then, the Softmax function computes the classification probabilities from the output vectors, the maximum of which is considered as the classification result.

Algorithm 1. The Learning Rate and Parameter Update in the Proposed DBGC-ATFFNet-AFTL Method

Input: Raw EEG signals x , the corresponding class labels y , the maximum epoch τ , learning rate ξ , regularization weight λ , the DBGC-ATFFNet-AFTL $Net(\cdot)$;

Output: The affective computing result O ;

```

1 Initializing parameters in the as  $\theta^{(0)}$ ;
2 Initializing data  $D_i$  in one batch with  $i = 1, 2, \dots, N$ ;
3 Initializing  $\tau = 200, \xi = 1 \times 10^{-3}, q = 0$ ;
4 while  $q \neq \tau$  do
5   for  $i = 1$  to  $N$  do
6     Generate conditional probability  $p_j = Net(\theta^{(q)}, D_i)$ ;
7     Calculate loss  $J^{(q)}$  on  $x$  by Eq. (8);
8     Calculate the gradient  $g = \nabla J^{(q)}$ ;
9     Update the parameters:  $\theta^{(q+1)} \leftarrow \theta^{(q)} - \xi \times g$ ;
10  end for
11   $q++$ ;
12 end while
13 Get the classification result  $O = Net(\theta^{(\tau)}, x)$ ;
```

To summarize, the optimizing procedure of our proposed DBGC-ATFFNet-AFTL is shown in Algorithm 1. The model is trained by minimizing the cross-entropy loss J between model prediction and the label, which is defined by:

$$J = \sum_{j=1}^N \sum_{i=1}^M -\log(p_i) \omega(y_i = l_i) + \lambda \|\theta\| \quad (8)$$

where p_i is the j -th conditional probability generated by the model, l_i is the j -th class from the label set L , $\omega(\cdot)$ is the

TABLE 1
Details of SEED, SEED IV, and DEAP

Item	SEED	SEED-IV	DEAP
Channel Num	62	62	32
Subject Num	15	15	32
Video Num	15	24	40
Stimulus Materials	Film Clips	Film Clips	Music Videos
Emotion Status	3 classes	4 classes	—

indicator function. In order to avoid the overfitting problem of the proposed model, we also introduce the trade-off regularization to the Eq. (8), where θ refers to the learnable parameters of the model and λ is the regularization weight.

3 EXPERIMENTAL RESULTS

3.1 EEG Datasets

The effectiveness of the proposed DBGC-ATFFNet-AFTL is evaluated on three public EEG datasets which are described in this section:

- 1) *SJTU Emotion EEG Dataset*: (SEED) [12] contains EEG data of 15 subjects (7 males and 8 females), which are collected via 62 EEG electrodes from the subjects when they are watching fifteen Chinese film clips with three types of emotions, i.e., negative, positive and neutral. Each subject has three sessions and there are 15 trails (5 for each class) per session. The EEG signals were recorded and down sampled to 200 Hz by 62 electrodes.
- 2) *SJTU Emotion EEG Dataset IV*: (SEED IV) [25] comprises EEG data of 15 subjects (7 males and 8 females) recorded in 62 channels. The experiment setting is the same as SEED. The data were collected when participants watch movies in four types of emotions, namely neutral, sad, fear, and happy, and the eye movement features are not used in this paper. Each movie lasts around 2 minutes. Three sessions of data are collected and each session comprises 24 trials/movies for each subject.
- 3) *Database for Emotion Analysis using Physiological Signals*: (DEAP) [19] consists of EEG data from 32 subjects (16 males and 16 females) who were shown 40 music videos, and the physiological signals of the subjects were recorded. Additionally, the subjects specified rating values according to four emotional states (valence, arousal, liking, and dominance) using consecutive numbers between 1 and 9. The length of the DEAP data was 63s sampled at 128 Hz in 32 channels.

As is discussed in [26], there are repeated sessions in SEED and the first one reflects stronger emotion feedback that is more reliable than the latter two sessions, we also only use the first session for each subject to ensure the consistency of evaluation in the experiment of this paper. We strictly follow the evaluation protocol in [5] for all three datasets and the detail information of three datasets can be found in Table 1.

3.2 Evaluation Metrics and Models

The proposed DBGC-ATFFNet-AFTL is evaluated by the classification accuracy (Acc) [12], F1-score (F1) [27], and

TABLE 2
The Overall Comparison of Classification Performance on SEED Dataset

subject	DGCNN [17]			4D-CRNN [18]			resHGCN [27]			Ours		
	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC
1	89.39 $\pm$ 0.93	0.892	0.964	93.75 $\pm$ 0.93	0.936	0.980	95.22 $\pm$ 1.27	0.941	0.984	97.41 $\pm$ 2.46	0.973	0.994
2	80.00 $\pm$ 1.95	0.799	0.920	87.97 $\pm$ 0.71	0.879	0.965	87.83 $\pm$ 1.97	0.867	0.952	94.81 $\pm$ 0.79	0.947	0.984
3	83.85 $\pm$ 1.34	0.838	0.936	91.69 $\pm$ 1.11	0.916	0.975	91.74 $\pm$ 1.97	0.907	0.971	95.93 $\pm$ 0.65	0.959	0.990
4	94.01 $\pm$ 1.87	0.939	0.986	97.05 $\pm$ 1.15	0.970	0.992	98.23 $\pm$ 0.66	0.972	0.995	98.05 $\pm$ 1.01	0.980	0.996
5	85.12 $\pm$ 1.09	0.850	0.950	92.33 $\pm$ 1.23	0.923	0.969	93.28 $\pm$ 1.54	0.922	0.978	97.49 $\pm$ 1.02	0.974	0.993
6	91.45 $\pm$ 0.78	0.913	0.970	94.84 $\pm$ 1.28	0.948	0.989	97.40 $\pm$ 0.49	0.963	0.991	98.08 $\pm$ 0.66	0.980	0.995
7	91.45 $\pm$ 1.41	0.913	0.968	94.66 $\pm$ 0.66	0.946	0.984	95.72 $\pm$ 1.00	0.946	0.981	97.96 $\pm$ 0.72	0.979	0.995
8	87.77 $\pm$ 1.74	0.876	0.958	92.54 $\pm$ 1.08	0.924	0.982	93.72 $\pm$ 0.95	0.927	0.979	95.82 $\pm$ 0.31	0.957	0.991
9	94.37 $\pm$ 1.66	0.943	0.985	97.14 $\pm$ 0.81	0.971	0.990	97.93 $\pm$ 0.72	0.979	0.995	97.19 $\pm$ 0.56	0.961	0.990
10	82.99 $\pm$ 1.50	0.827	0.928	93.75 $\pm$ 1.29	0.936	0.979	93.95 $\pm$ 1.60	0.929	0.974	95.43 $\pm$ 5.60	0.953	0.990
11	92.10 $\pm$ 0.67	0.920	0.979	94.52 $\pm$ 1.39	0.945	0.986	96.05 $\pm$ 1.15	0.950	0.989	97.58 $\pm$ 0.55	0.976	0.995
12	90.04 $\pm$ 1.99	0.900	0.970	95.63 $\pm$ 0.71	0.956	0.989	95.99 $\pm$ 1.04	0.949	0.986	98.47 $\pm$ 0.31	0.984	0.997
13	90.66 $\pm$ 1.04	0.905	0.961	95.25 $\pm$ 1.69	0.952	0.985	95.34 $\pm$ 0.70	0.942	0.981	97.52 $\pm$ 0.49	0.975	0.995
14	92.60 $\pm$ 1.25	0.925	0.982	96.05 $\pm$ 0.72	0.960	0.991	98.52 $\pm$ 0.68	0.975	0.995	98.70 $\pm$ 0.35	0.987	0.997
15	97.79 $\pm$ 0.46	0.977	0.996	99.44 $\pm$ 0.48	0.994	0.999	99.46 $\pm$ 0.22	0.994	0.999	99.23 $\pm$ 0.36	0.992	0.999
Aver	89.39 $\pm$ 0.93	0.892	0.964	94.44 $\pm$ 2.70	0.944	0.984	95.38 $\pm$ 3.06	0.943	0.983	97.31 $\pm$ 1.47	0.972	0.993

Where bold fonts indicate best results.

area under curve (AUC) [28], respectively. Moreover, standard deviation (Std) [29] is introduced to assess the robustness of our proposed model. Specifically, for a certain subject, the Std value is calculated based on the accuracy in multiple folds of testing, while for the average result, Std is obtained from average accuracies of all subjects. In order to compare with the proposed method, we use three baseline models and all these models are retested on three datasets in the same experiment environment. We faithfully reproduced these deep learning models and brief introductions of the three baseline models are given as follows:

- 1) *DGCNN* [17]: This method uses a dynamic adjacent matrix to simulate the channel relationship with shallow DE features, which has shown its ability in EEG feature extraction.
- 2) *4D-CRNN* [18]: This is a CNN-based model which also combines recurrent neural network and DE features to integrate both spatial and temporal information.
- 3) *resHGCN* [27]: This is a deep learning model with the residual architecture and an encoded adjacent matrix using multi fully connected layers, which probably retains DE features of preprocessed EEG signals.

Moreover, for further investigating the importance of each feature in our DBGC-ATFFNet-AFTL, we carry out the ablation study with two simplified models, each of which consists of only one feature branch.

3.3 Performance of Subject-Dependent Experiments

In order to evaluate the efficacy of the proposed framework, we compare our DBGC-ATFFNet-AFTL with the above baseline models. Tables 2 and 3 list the classification results of subject-dependent experiments on SEED and SEED IV datasets. From Table 2 we can learn that all three baseline models can achieve relatively high average accuracies on 15 subjects of SEED dataset. Especially, our proposed DBGC-ATFFNet-AFTL has the highest average Acc, F1, and AUC. In terms of Acc, our DBGC-ATFFNet-AFTL reaches 97.31%,

which is 1.93%, 2.87%, and 7.92% higher than baseline methods respectively. The obvious improvement on Acc shows that our proposed method can better fuse two types of features and in the meantime, a relatively low standard deviation of our method indicates that its robust ability of high-level feature extraction.

In terms of SEED IV dataset, our proposed approach gets 89.97%, 0.898 and 0.971 on Acc, F1, and AUC, which performs better than all three other baseline models. Moreover, a relatively low standard deviation of 2.85% demonstrates the robustness of our proposed method. Particularly, since there are totally 32 subjects in DEAP dataset, we summarize the experiment results in Fig. 4 due to space limitation. The confusion matrixes are utilized for validation in Fig. 5. We can find that our proposed DBGC-ATFFNet-AFTL can achieve encouraging results of VA-space (4-class: HVHA, HVLA, LVHA, LVLA) classification on Acc, F1 and AUC metrics. We can see that our DBGC-ATFFNet gains outstanding accuracy results on all three datasets, with an impressive recognition performance close to 1 in three kinds of emotions on SEED dataset. As a result, the above experimental results prove the ability of our proposed method in EEG emotion classification.

To demonstrate the superiority of our DBGC-ATFFNet-AFTL on emotion recognition, Table 4 makes a comparison between classification results reported in recent years on both SEED and SEED IV datasets. Since most methods only report Acc and Std on their papers, we use these two metrics for comparison here. From Table 4, we can learn that our proposed method has a better classification performance than most of these methods for the SEED and SEED IV datasets. Compared with the methods which adopted the single DE feature with GNNs, our method fuses both the DE and PSD features and gains accuracy increases of 1.21% and 7.24% on the subject-dependent task of SEED and SEED IV datasets. Moreover, in comparison with 4D-CRNN which used convolution operation, our proposed method with Adaptive Transformer cares not only the spatial location relationship but also the global connection of

TABLE 3
The Overall Comparison of Classification Performance on SEED IV Dataset

Subject	DGCNN [17]			4D-CRNN [18]			resHGCN [27]			Ours		
	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC	Acc $\pm$ Std(%)	F1	AUC
1	69.44 $\pm$ 5.37	0.684	0.911	83.24 $\pm$ 4.05	0.841	0.949	86.24 $\pm$ 4.86	0.862	0.951	87.84 $\pm$ 3.43	0.878	0.961
2	91.55 $\pm$ 9.17	0.914	0.985	90.42 $\pm$ 1.20	0.903	0.977	90.59 $\pm$ 1.11	0.906	0.974	90.66 $\pm$ 2.17	0.907	0.974
3	92.71 $\pm$ 1.75	0.925	0.981	87.60 $\pm$ 2.36	0.876	0.955	87.31 $\pm$ 1.95	0.873	0.961	89.07 $\pm$ 1.11	0.896	0.975
4	94.24 $\pm$ 2.85	0.943	0.987	89.95 $\pm$ 4.25	0.899	0.972	91.56 $\pm$ 4.15	0.909	0.977	91.54 $\pm$ 3.02	0.914	0.976
5	79.31 $\pm$ 8.26	0.791	0.947	85.48 $\pm$ 2.82	0.850	0.954	86.13 $\pm$ 2.06	0.857	0.956	87.48 $\pm$ 1.47	0.871	0.963
6	85.44 $\pm$ 8.28	0.854	0.967	88.19 $\pm$ 2.99	0.881	0.970	89.31 $\pm$ 1.87	0.891	0.971	90.43 $\pm$ 2.77	0.903	0.969
7	78.48 $\pm$ 16.19	0.768	0.959	86.42 $\pm$ 2.62	0.861	0.955	88.01 $\pm$ 2.74	0.883	0.933	90.02 $\pm$ 4.03	0.900	0.969
8	88.83 $\pm$ 3.78	0.886	0.973	87.95 $\pm$ 3.82	0.878	0.953	88.36 $\pm$ 3.32	0.884	0.962	91.42 $\pm$ 1.35	0.911	0.974
9	92.22 $\pm$ 2.69	0.922	0.979	90.78 $\pm$ 2.23	0.906	0.979	91.89 $\pm$ 1.83	0.917	0.980	91.77 $\pm$ 4.13	0.915	0.982
10	80.25 $\pm$ 11.25	0.801	0.940	87.71 $\pm$ 2.06	0.875	0.958	88.36 $\pm$ 4.11	0.879	0.965	92.48 $\pm$ 1.66	0.923	0.973
11	65.45 $\pm$ 11.60	0.602	0.919	85.25 $\pm$ 3.47	0.850	0.950	84.95 $\pm$ 3.03	0.848	0.948	85.84 $\pm$ 1.85	0.854	0.957
12	73.22 $\pm$ 17.06	0.703	0.934	83.72 $\pm$ 1.11	0.835	0.946	82.61 $\pm$ 3.34	0.825	0.946	84.14 $\pm$ 2.91	0.838	0.956
13	86.48 $\pm$ 6.24	0.863	0.971	88.77 $\pm$ 2.34	0.887	0.963	89.42 $\pm$ 2.15	0.896	0.970	90.83 $\pm$ 1.94	0.905	0.973
14	82.83 $\pm$ 11.98	0.828	0.957	86.26 $\pm$ 2.62	0.862	0.958	91.89 $\pm$ 1.28	0.918	0.970	89.54 $\pm$ 2.68	0.891	0.965
15	96.70 $\pm$ 1.59	0.966	0.991	93.71 $\pm$ 2.06	0.935	0.987	95.65 $\pm$ 3.21	0.955	0.961	96.59 $\pm$ 1.82	0.961	0.992
Aver	83.81 $\pm$ 9.41	0.830	0.960	87.75 $\pm$ 2.72	0.876	0.962	88.79 $\pm$ 3.21	0.886	0.966	89.97 $\pm$ 2.85	0.898	0.971

Where bold fonts indicate best results.

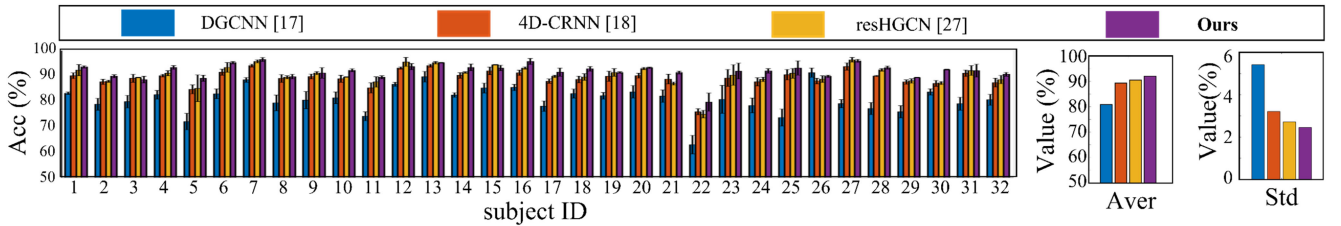


Fig. 4. The overall comparison of classification performance on DEAP dataset.

different EEG channels, and reaches 3.32% higher in accuracy. Especially for SEED IV dataset, which contains shorter records of EEG signals, our method learns the common pattern of feature distribution from the subjects and has a more robust classification performance. Additionally, the performance comparison on three kinds of classification tasks of DEAP dataset is listed in Table 5, and our proposed method performs better on Valence, Arousal and VA-Space classification tasks and gains relatively lower standard deviation of 3.27%, 2.89% and 3.10% respectively. Consequently, it is obvious that our DBGC-ATFFNet-AFTL can distinguish between EEG signals of different emotions more accurately and effectively.

3.4 Performance of Cross-Subject Experiments

Apart from encouraging results in the subject-dependent experiment, our proposed DBGC-ATFFNet-AFTL also has

an impressive classification performance in cross-subject experiments. As DEAP is a dimensional dataset [19] and there was no previous work which adopted this dataset for similar testing, the experiments are carried out on SEED and SEED IV datasets for further comparison. There are totally 15 subjects in each dataset, to apply adapter-fine-tuned transfer learning algorithm, we split the whole dataset into two parts: EEG signal data of 14 subjects is used for pretraining, and EEG data of the rest one subject is used for finetuning and evaluating the performance of the proposed method. The detailed process is depicted in Fig. 1

Firstly, our proposed approach is pretrained on the source data and then freeze all the learnable parameters except the Adapter modules. It is proven through a series of

TABLE 4
Comparison with the State-Of-The-Art Methods of Subject-Dependent on SEED and SEED IV Datasets

Methods	Year	SEED	SEED IV
		Acc $\pm$ Std (%)	Acc $\pm$ Std (%)
GELM [5]	2017	91.07 $\pm$ 7.54	-
DGCNN [17]	2018	90.40 $\pm$ 8.49	-
4D-CRNN [18]	2020	94.74 $\pm$ 2.32	-
RGNN [30]	2020	94.24 $\pm$ 5.95	79.37 $\pm$ 10.54
3DCNN&PST [31]	2021	95.76 $\pm$ 4.98	82.73 $\pm$ 8.96
4D-aNN [14]	2022	96.10 $\pm$ 2.61	-
Ours	2022	97.31 $\pm$ 1.47	89.97 $\pm$ 2.85

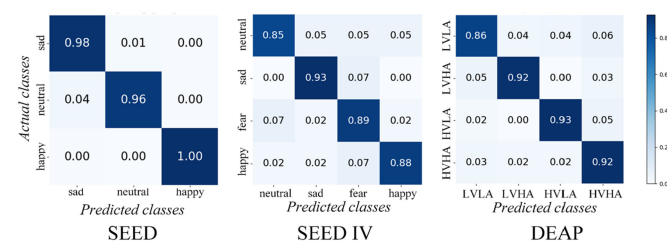


Fig. 5. The confusion matrixes of our proposed method on all three datasets.

Where bold fonts indicate best results.

TABLE 5
Comparison With the State-of-the-Art Methods of Subject-Dependent on DEAP Dataset

Methods	Year	Valence	Arousal	VA-Space
		Acc $\pm$ Std(%)	Acc $\pm$ Std(%)	Acc $\pm$ Std(%)
GELM[5]	2017	-	-	69.97
DGCNN[17]	2018	86.23 $\pm$ 12.29	84.54 $\pm$ 10.18	85.02 $\pm$ 10.25
MMResLSTM[32]	2019	92.31 $\pm$ 1.55	92.87 $\pm$ 2.11	-
4D-CRNN[18]	2020	93.52 $\pm$ 3.26	93.78 $\pm$ 4.19	-
ACRNN[33]	2020	93.72 $\pm$ 3.21	93.38 $\pm$ 3.73	-
SparseDGCNN[26]	2021	95.72 $\pm$ 3.75	91.75 $\pm$ 5.23	-
Ours	2022	95.91 $\pm$ 3.27	94.61 $\pm$ 2.89	91.98 $\pm$ 3.10

Where bold fonts indicate best results.

experiments that our adapter-finetuned transfer learning strategy has the optimal classification performance with a relatively low resource of data when 50 percent of the target samples are randomly chosen to finetune the parameters of Adapters. Based on the well-trained model, finally, we carry out the evaluation experiments on the rest EEG data of the target subject. To demonstrate the superiority of our Adapter-finetuning method on cross-subject emotion recognition, Table 6 makes comparison between classification results reported in recent years on both SEED and SEED IV datasets. Our method gains accuracy of 94.39% and 89.78% on SEED and SEED IV datasets respectively. Especially for SEED IV dataset, which contains shorter records of EEG signals, our proposed method learns the common pattern of feature distribution from source subjects and has a more robust classification performance on the target subject.

4 DISCUSSIONS

4.1 Efficacy of Dual-Branch Dynamic Graph Convolution Network

To validate the superiority of our extracted features, the t-SNE visualization is utilized. The t-SNE visualizes the extracted EEG features into a 2D embedding space. The experimental results for all three datasets are presented in Fig. 6. From the t-SNE plots, compared with those methods that applied DGCNN [17], 4D-CRNN [18] and resHGCN [27] to single DE features, our proposed method obtains an outstanding classification performance with considerable inter-class distance on discrete emotion datasets: SEED and SEED IV, while there is a relatively dense distribution of features for the DEAP, which is a dimension emotion

dataset. It is obvious that the introduction of PSD features compensates for the lack of information in the spectral domain of the EEG signals. Additionally, we convolve these two types of pre-computed features with dual branches of graph convolution to make the most of different distributions in temporal and spectral domain. This enhancement indicates that it is helpful to input more pre-processed features and there is probably complementary information in these two types of features, which are both beneficial to the extraction of high-level features.

4.2 Efficacy of the Adaptive Transformer Feature Fusion Network

To assess the effectiveness of Adaptive Transformer Feature Fusion Network, we conduct ablation experiments. From the results which are listed Table 7, we can find that accuracy increases from 95.44%, 88.75% and 90.46% to 97.31%,

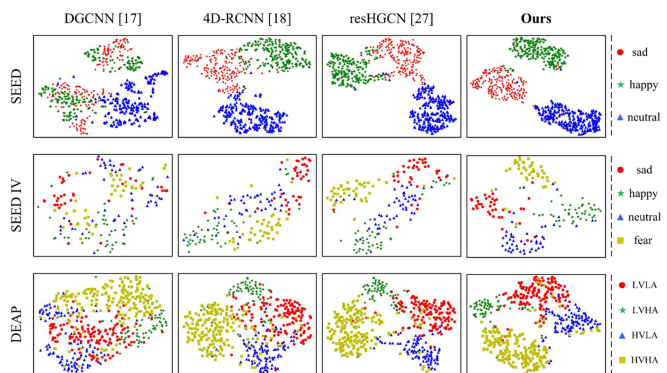


Fig. 6. The t-SNE visualization in 2D embedding space of different types of features on all three datasets.

TABLE 6
Comparison With the State-of-the-art Methods of Cross-Subject on SEED and SEED IV Datasets

Method	Year	SEED	SEED-IV
		Acc $\pm$ Std (%)	Acc $\pm$ Std (%)
DGCNN[17]	2018	79.95 $\pm$ 9.02	—
RGNN[30]	2020	85.30 $\pm$ 6.72	73.84 $\pm$ 8.02
TANN[34]	2021	84.41 $\pm$ 8.75	68.00 $\pm$ 8.35
BiHDM[35]	2021	85.40 $\pm$ 7.53	69.03 $\pm$ 8.66
Ours	2022	94.39 $\pm$ 3.23	89.78 $\pm$ 3.09

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TABLE 7
Ablation Study of Adaptive Transformer Feature Fusion Network on All Three Datasets

Method	SEED	SEED IV	DEAP
	Acc $\pm$ Std (%)	Acc $\pm$ Std (%)	Acc $\pm$ Std (%)
w / o ATFFNet	95.44 $\pm$ 3.06	88.75 $\pm$ 3.72	90.46 $\pm$ 3.98
w / ATFFNet	97.31$\pm$1.47	89.97$\pm$2.85	91.98$\pm$3.10

Where bold fonts indicate best results, “w / o” denotes “without” and “w /” denotes “with”.

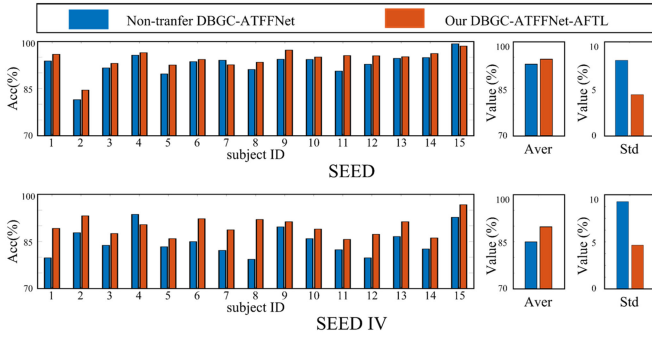


Fig. 7. Comparison between non-transfer DBGC-ATFFNet and our DBGC-ATFFNet-AFTL on SEED and SEED IV datasets.

89.97%, and 91.98% on all three datasets with relatively low standard deviations. In general, the outstanding performance of our DBGC-ATFFNet-AFTL method indicates that ATFFNet unit effectively fuses the foremost extracted features and explores the connection between EEG channels more comprehensively in the meantime. Unsimilar to 2D convolution and 3D convolution, which are commonly used in previous works [18], the Adaptive Transformer in our ATFFNet can avoid the overdependence on the explicit spatial location distribution. Moreover, the features with the sequential form are beneficial to decrease the number of trainable parameters of the model and reduce the risk of overfitting. Additionally, the Channel-Weight unit reinforces the learning of the model about the channel connection with the existing trainable parameters.

4.3 Efficacy of the Adapter-Finetuned Transfer Learning

In order to assess the effectiveness of our adapter-finetuned transfer learning method, we apply the same protocol to the corresponding subject-dependent experiments and the results of two training methods on SEED and SEED IV datasets are compared in Fig. 7 and Table 8, respectively. We explore the effectiveness of the Adapter-finetuning method from two perspectives. In terms of the classification performance, from Fig. 7, we can learn that the proposed DBGC-ATFFNet-AFTL with finetuned Adapters reaches average accuracies of 94.39% and 89.78% on two datasets, which are both higher than the non-transfer DBGC-ATFFNet method. Especially on SEED IV dataset, which has a relatively small number of EEG samples, the accuracy of the model using transfer learning is 4.81% higher than the non-transfer one. The experiment results indicate the strategy of transfer learning indeed helps the model learn to extract higher-level features which are easily missed in analyzing a single subject. Thus, it is obvious that transfer learning based on

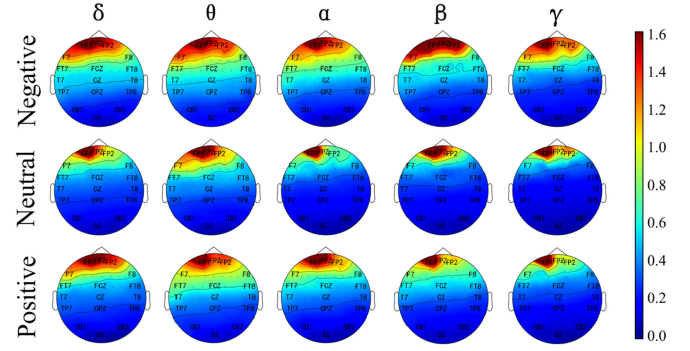


Fig. 8. Saliency maps of the DBGC-ATFFNet-AFTL on SEED dataset.

finetuning Adapter module contributes to EEG emotion classification of crossing subjects. In terms of computational cost, as in Table 8, our proposed Adapter-finetuned DBGC-ATFFNet spends less time for training and has less parameters involved, which enhances the efficiency of transfer learning. It is worth mentioning that the training and testing of the deep learning model are performed on Nvidia Tesla V100 GPU with 32 GB memory. In this experiment environment, it averagely just takes 53.3s and 22.2s to finish the finetuning step and apply the pretrained model to the target subject on SEED and SEED IV datasets respectively, which also significantly reduces the quantity of the trainable parameters in the meantime compared with the non-transfer model.

4.4 Saliency Map Analysis of EEG Channels

In order to demonstrate the interpretability of our proposed approach, we apply the saliency map method [36], [37] based on the gradient propagation, which is widely used in the computer vision area. Fig. 8 depicts the average neural patterns for positive, neutral and negative emotions in five frequency bands. The visualization on the scalp maps presents spatial distributions for emotion recognition tasks, directly reflecting how our proposed method behaves when making inferences on EEG signals. For example, from Fig. 8, the emotionally active areas are mainly concentrated on the left prefrontal and parietal sites. Brain activities associated with emotions generally have a significant response in β bands. The findings of these saliency maps have been demonstrated and are in line with the existing emotion studies [38], [39], [40]. Apart from these findings, we can further learn that the responses of both the β and γ bands are more notable in all three kinds of emotional states. For neutral emotions, the neural patterns tend to be gentler compared with the positive and negative, while for negative emotions, there

TABLE 8
Comparison of Parameters and Computational Cost on SEED and SEED IV Datasets

Dataset	Method	Parameter (thousand)	Time Cost(s)
SEED	Non-transfer DBGC-ATFFNet	241.239	398.6
	Our DBGC-ATFFNet-AFTL	1.456	53.3
SEED-IV	Non-transfer DBGC-ATFFNet	237.540	156.7
	Our DBGC-ATFFNet-AFTL	1.456	22.2

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are significantly higher β responses at the prefrontal cortex, and the lateral temporal and parietal areas activate more, which are similar to those of positive emotions.

4.5 Limitations and Future Directions

Although the proposed method achieves outstanding classification results, our present work still suffers from several limitations. First, the extracted features from preprocessed EEG signals are constraint to prior knowledge, which may lead to loss of some useful information hidden in original EEG signals. Therefore, our important future work is to handle affective computing tasks with different types of feature maps, which are directly extracted from raw signals by using end-to-end deep learning models. Second, although our method shows the effectiveness in cross-subject emotion classification, we involve all the available EEG channels into transfer learning. It is possible that several channels make greater contribution to affective computing while others provide noisy signals that are harmful to the analysis. Thus, we will consider adaptively selecting a few EEG channels with outstanding performance to improve our method in the future work.

5 CONCLUSION

In this article, we propose a novel dual-branch dynamic graph convolution based adaptive Transformer feature fusion network (DBGC-ATFFNet-AFTL) for EEG emotion recognition. Specifically, our proposed DBGC-ATFFNet-AFTL first compute differential entropy and power spectral density features from preprocessed EEG signals. Next, the dual-branch dynamic graph convolution network extracts high-level EEG information of different emotions through two branches of feature inputs. Moreover, the adaptive transformer feature fusion network is further employed to fuse the obtained features and learn the connection relationship of EEG channels simultaneously in the meantime. Additionally, we introduce adapter-finetuned transfer learning to the cross-subject emotion classification task by finetuning the Adapter modules. We conduct experiments on three public emotional EEG datasets to evaluate the effectiveness of our proposed DBGC-ATFFNet-AFTL method. Experimental results show that our proposed method has a better performance in accuracy, F1-score and AUC value. Detailed discussions demonstrate our proposed approach is able to effectively learn emotion information from EEG signals, which has a potential application prospect in the affective computing area.

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Mingyi Sun received the bachelor's degree in engineering from Beihang University, Beijing, China, in 2021, where he is currently working toward the master's degree with the Department of Automation Science and Electrical Engineering. His current research interests include signal processing, machine learning, and brain-computer interface.



Weigang Cui received the bachelor's degree in mathematics and PhD degree from the Department of Automation Science and Electrical Engineering, Beihang University, Beijing, China, in 2016 and 2021, respectively. He is currently doing post-doctoral research with the School of Engineering Medicine, Beihang University. His research interests include medical image analysis, machine learning, and brain functional connectivity.



Shuyue Yu received the BS degree in measurement, control technology, and instrument and the MS degree in control science and engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2016 and 2019, respectively. She is currently an engineer with Beijing Aerospace Measurement and Control Technology Company, Ltd. Her current research interests include robotics and BCI.



Hongbin Han received the undergraduate and graduate degrees in clinical medicine from Dalian Medical University, Dalian, China, in 1996, and the doctor of radiology degree from Peking University Health Center, Beijing, China, in 1998. He completed the radiology residency with the Peking University Health Center, joined the faculty of the Department of Radiology, Peking University Third Hospital in 1998, and is currently the professor with the Radiology Department. He has established the Beijing MRI Technology Research Laboratory and severed as the Director since 2010. His research interests include the development and application of advanced MRI techniques for diagnosis, and therapy of human brain diseases.



Bin Hu (Member, IEEE) received PhD degree in computer science from the Institute of Computing Technology, Chinese Academy of Science, China, in 1998. Since 2008, he has been a professor with the School of Information Science and Engineering, Lanzhou University, China. He had been also guest professorship in ETH Zurich, Switzerland till 2011. His research interests include pervasive computing, computational psychophysiology, and data modeling.



Yang Li received the PhD degree in automatic control and systems engineering from the University of Sheffield, Sheffield, U.K., in 2011. He did post-doctoral research with the Department of Computer and Biomedical Engineering, University of North Carolina at Chapel Hill, Chapel Hill, NC, for one year. In 2013, he joined the Department of Automation Sciences and Electrical Engineering, Beihang University, Beijing, China, as a professor. His current research interests include system identification and modeling for complex nonlinear processes: NARMAX methodology and applications, nonstationary signal processing and sparse representation, medical image analysis, and brain-computer interface.

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